

13	B	Maximum kinetic energy of the air molecule $E_0 = \frac{1}{2}mv_0^2$ $= \frac{1}{2}m(x_0\omega)^2$ $= \frac{1}{2}\left(\frac{2.9 \times 10^{-2}}{6.02 \times 10^{23}}\right)\left(2.0 \times 10^{-6} \times \frac{2\pi}{3.0 \times 10^{-3}}\right)^2$ $= 4.226 \times 10^{-31}$ $\approx 4.2 \times 10^{-31} \text{ J}$
14	D	Periods of waves X and Y each occupy 3 large squares on the time axis